

NAYEEM HABIB

Srinager, Jammu and Kashmir

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Education

NIT, TIRUCHIRAPPALLI

Aug 2019 – Jun 2023

B.Tech - Electrical and Electronics Engineering

CGPA: 7.2/10

Skills

Programming Languages and Database: C/C++, Python, Verilog, VHDL, JavaScript, CSS, MySQL and MongoDB.

Technologies & Tools: VS Code, Xilinx Vivado, LTSpice, Matlab, Cadence Virtuoso, Simulink, Git and GitHub.

Frameworks and Libraries: React, Next.js, Node.js, Express.js, Django and Longhain.

Work Experience

UPSC Civil Services Preparation.

Jun 2023 - Dec 2025

Career Break – UPSC Preparation

- Prepared for the UPSC exam with **Electrical Engineering (EEE)** as my optional subject.
- Strengthened core concepts in **Power Systems, Control Systems, Machines, and Analog and Digital Electronics.**
- Developed strong analytical, time management, and self-discipline skills.

IIT Madras, Tamil Nadu.

May 2022 - Jul 2022

Research Internship : [Link](#)

- Implemented the Particle Swarm Optimization algorithm in Verilog, and Compared the final Result with the Matlab.
- Analyze three Verilog modules to display a specific **16-bit** floating point number on the FPGA board.
- Used **Matlab, C++, Verilog and FPGA Board.**

NIT Tiruchirapalli, Tamil Nadu.

May 2021 - Jul 2021

Research Internship : [Link](#)

- Developed a MultiFace LIPSYN Simulation Model using deep learning.
- Used Speaker Diarization, Face Detection and Recognition, Lip Synchronization Smoothing and Super Resolution using **GFPGAN.**
- Used **DeepFilterNet, CNN, TensorFlow, Keras, PyTorch and NumPy.**

Project Work

RLC Filter Circuit Design and Analysis : [GitHub](#)

March 2023 - April 2023

- Developed a simulation model of an RLC filter circuit using Python, achieving an **accuracy of up to 98 Percent.**
- Analyzed over **100 configurations** of RLC circuits to identify the optimal configuration. Used **Python, NumPy, Matplotlib and SciPy.**

Digital Signal Sampling and Analysis : [GitHub](#)

June 2022 - July 2022

- Simulated DSP concepts such as **sampling, aliasing, DFT, and signal** reconstruction using MATLAB. Analyzed spectral resolution, frequency spectrum, and interpolation techniques for discrete-time signals.
- Validated **Nyquist-Shannon sampling** theory through practical signal processing experiments.
- Used **Python, NumPy, Matplotlib and SciPy.**

Circuit Delay Prediction for CMOS Timing Analysis : [GitHub](#)

Jan 2022 - Feb 2022

- Developed a simulation model of traffic light control system using AI with **accuracy to 99 Percent.** Utilized computer vision techniques to determine time limits on roads. Used Python, Image Processing, Object Detection and TensorFlow.

Face Recognition Biometric System : [GitHub](#)

Nov 2021 - Dec 2021

- Developed a simulation model of face recognition biometric system. Implemented it over the **60 Students** in Class room.
- Used Python, Image Processing, Image Recognition, OpenCV, SciPy, SciKit.

Vehicle Tracking System : [Link](#)

Aug 2021 - Sep 2021

- Developed a GPS tracker using Arduino to transmit location data to mobile devices via a GSM module; successfully tested on more than **10 devices**, with reliable performance over a **distance of up to 1 km.**
- Used Arduino UNO, GSM Module, GPS Module, Power Supply, Connecting Wires.

TRAFFIC LIGHT MANAGEMENT USING AI : [GitHub](#)

Mar 2021 - Apr 2021

- Developed a Simulation model of traffic light control system using. AI With accuracy to **99 Percent**, Utilized computer vision techniques to determine time limits of Vehicles on roads.
- Used **Python, Image Processing, Object Detection and TensorFlow..**

Achievements and Certificates

- **Competitive Programming** : Specialist on Codeforces (**max 2061**) and Knight on LeetCode (**max 1979**).
- Geo-Data Sharing and **Cyber Security (ISRO)** and Advanced Learning Algorithms From Coursera.

Responsibilities

- Team Leader in Analog Domain **Tronix 3D Aeromodelling** Club NIT Trichy . President in Current Events NIT Trichy. Workshop manager for EEE Association's NIT Trichy . Class representative for **3rd and 4th year**, NIT Trichy.